MATH50003 Numerical Analysis

V.3 Discrete Convolutions

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Chapter V

Numerical Fourier series

1. Fourier Expansions and approximating Fourier coefficients
2. Discrete Fourier Transforms and interpolation
3. Convolutions via the DFT

How to capture translational invariance?

Number recognition should return same answer for shifted images

$f($ $) = f($ $) = f($ $)$

V.3.1 Matrices that commute with translations

Circulant matrices define discrete convolutions

Definition 36 (circulant matrix)**.**

$$C[\boldsymbol{a}] := \begin{bmatrix} a_0 & a_{n-1} & \cdots & a_2 & a_1 \\ a_1 & a_0 & a_{n-1} & \cdots & \vdots \\ \vdots & a_1 & a_0 & \ddots & \vdots \\ a_{n-2} & \cdots & \ddots & \ddots & a_{n-1} \\ a_{n-1} & a_{n-2} & \cdots & a_1 & a_0 \end{bmatrix}$$

Lemma 9 (operators which commute with translations)**.** *A matrix* $A \in \mathbb{C}^{n \times n}$ *satisfies* $AS = SA$ *if and only if it is a circulant matrix.*

Definition 37 (discrete convolution)**.** The *discrete convolution* of two vectors $\boldsymbol{a}, \boldsymbol{f} \in \mathbb{C}^n$ is given by

$$\boldsymbol{a} \star \boldsymbol{f} := C[\boldsymbol{a}]\boldsymbol{f}.$$

V.3.2 Diagonalising Convolutions

The DFT diagonalises just like Fourier transforms

Proposition 16 (diagonalising S)**.**

$$S = Q_n^\star \begin{bmatrix} 1 & & & \\ & \omega^{-1} & & \\ & & \ddots & \\ & & & \omega^{1-n} \end{bmatrix} Q_n$$

where Q_n is the DFT matrix and $\omega = \mathrm{e}^{2\pi \mathrm{i}/n}$.

Theorem 16 (diagonalising circulants)**.**

$$C[\boldsymbol{a}] = \sqrt{n} Q_n^\star \mathrm{diag}(Q_n \boldsymbol{a}) Q_n.$$